

Product Design Report

Project Title: Dual Mode Gym Equipment with Modular Load System (ClevBell)

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1. Introduction

In recent years, the demand for home and personal fitness equipment has seen significant growth, driven by the increasing emphasis on health and wellness. Modern gym-goers seek compact, versatile, and user-friendly machines that can accommodate multiple workout routines without occupying excessive space.

This report presents the product design of a dual-mode cable pulley, easy to set-up at home gym equipment that caters to these needs. The equipment integrates a vertical frame with linear guide rails, allowing users to perform exercises such as seated and standing rows (from a lower pulley) and lat pull down (from upper pulley).

Unlike conventional machines with fixed-weight stacks, this design allows users to utilize their existing circular disk-shaped dumbbells as weight resistances. The loading mechanism is designed to safely accommodate these weights, making the equipment both cost-effective and adaptable to individual strength levels.

2. Problem Definition

Background:

There is a growing need for versatile and space-efficient gym equipment that can cater to users seeking effective strength training without investing in multiple machines. Many available systems are either specialized for a single motion or come with built-in weight stacks, making them bulky, costly, and less flexible for users who already own free weights like dumbbell disks.

Identified Gap:

Most conventional gym machines lack modularity and require proprietary weight systems. Users with existing weight disks often find it difficult to integrate them into pulley-based training setups. Moreover, machines that offer both upward and downward resistance training in a single unit are rare, especially in compact, user-friendly formats.

Design Goal:

The aim is to develop a compact, dual-mode gym equipment that:

- Operates on linear guide rails to ensure controlled, stable motion.
- Allows users to attach standard circular dumbbell disks as variable loads.
- Supports two distinct modes of resistance training:
 - Pull-down mode using an overhead pulley mechanism.
 - Pull-up mode using a base-mounted pulley system.
- Is easy to manufacture, assemble, and maintain using common tools and materials.

Functional Requirements:

- Reliable and smooth linear motion in both operational modes.
- Quick and secure switching between pull-down and pull-up configurations.
- Safe and ergonomic handling during exercise.
- Compatibility with commonly available dumbbell disks of varying weights.

3. Need Identification and Benchmarking

Effective strength training equipment is essential for users looking to build muscle, improve posture, or recover from injuries. However, most multifunctional gym machines available in the market are either large, expensive, or designed around proprietary weight systems that cannot be easily adapted or expanded.

Through interaction with potential users and observation of existing home or basic gym setups, the following needs have been identified:

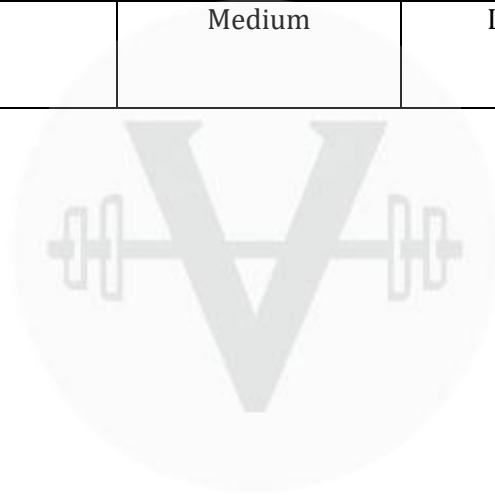
- **Modularity:** Users require a solution that allows flexibility in loading, ideally using weight disks they already possess (like dumbbell plates).
- **Versatility:** The machine should support a variety of workouts, particularly both pull-down and pull-up resistance-based exercises.
- **Compact Design:** The equipment must fit within limited spaces, such as home gyms or small workout areas.
- **Simplicity:** The equipment should be simple to use, adjust, assemble, and maintain with minimal instruction or training.
- **Safety and Reliability:** Smooth motion and secure loading mechanisms are critical to avoid injury and ensure a quality workout experience.

These insights point toward the need for a compact, dual-mode resistance machine that operates on a guided track and leverages the user's existing weight inventory.

Benchmarking:

To ensure the proposed design improves upon existing solutions, a comparison was made between various market options and the proposed equipment. Key parameters were identified:

Parameter	Commercial Lat Pull Down Machine	Home Gym Pulley	Adjustable Resistance Bands	Proposed Design
Cost	High	Low to Medium	Low	Medium
Weight Compatibility	Fixed internal weights	Partial	No	Yes
Dual Mode Functionality	No	No	No	Yes
Ease of Manufacturing	Complex	Moderate	Simple	Moderate
Motion Smoothness	High	Medium	Low	High



4. Concept Generation

The goal of this product design is to create a versatile gym machine that enables users to perform multiple upper body exercises using a compact and portable setup, while utilizing their existing dumbbell plates for resistance. The concept prioritizes space-efficiency, ease of use, and low cost, making it ideal for home workouts.

2.1 Functional Requirements

- Enable two distinct workout modes:
 - Lat Pull Down (via upper pulley)
 - Triceps' Pushdown / Low Row (via bottom pulley)
- Use of user-owned circular dumbbell plates as resistance
- Incorporate fixed pulleys (one at the top, one at the bottom)
- Allow quick rope repositioning between upper and lower pulleys
- Ensure smooth linear movement through guide rails
- Compact, foldable or collapsible frame for easy storage

2.2 Design Concept

After considering several configurations, a single compact design was selected that satisfies all functional needs without adding complexity. The key elements of the final concept are:

- A vertical frame with linear guide rails mounted on a base plate to ensure stability.
- A bar assembly connected to a pulley cable that moves along the guide rails.
- Three fixed pulleys, positioned at the top (2) and bottom (1) of the frame.
- A detachable cable system: the user can switch between lat pulldown and low pulley exercises by simply repositioning the cable (from top pulley to bottom and vice versa).
- A plate loading rod on weight holder that slides on the linear guide rails, where dumbbells can be securely attached.
- Compact storage design: the vertical frame can be made foldable or include telescopic rails to minimize space when not in use.

2.3 Rationale for Concept Selection

This concept was chosen due to its:

- Simple mechanical structure, minimizing moving parts and maintenance.
- User adaptability, allowing quick transitions between exercise modes.

- Space efficiency, suitable for home or small gym environments.
- Affordability, as it avoids built-in weight stacks and relies on user-owned dumbbells.

This finalized concept strikes a balance between functionality, and user convenience, and will be carried forward into the design and analysis phases.

5. Product Design Cycle Stages

A. Embodiment Design

The equipment is designed with a modular and durable structure that supports both pull-down and pull-up exercises. Each component has been chosen based on mechanical properties, ease of manufacturing, and suitability for the intended use.

- **Bar:** Made from HR Steel 1006, selected for its high tensile strength and ductility, making it ideal for repetitive loading in a gym setup.
- **Guide Rail Carriage:** Constructed from ASTM 20 gray cast iron, which offers excellent wear resistance and vibration damping, ensuring long-term smooth operation.
- **Bottom Frame:** Made from aluminum for its lightweight yet strong properties. The core frame is a rectangle of 45 cm × 60 cm. Two 45 cm sides extend further—one side to 60 cm and the other to 55 cm—creating an asymmetric base that enhances lateral stability.
- **Guide Rail Plate:** A 2-meter-long aluminum plate forms the vertical track for the linear carriage. It has pre-drilled holes at the top to mount a pulley shaft.
- **Pulley System:** Two pulleys are mounted at the top of the guide rail plate using a shaft made of HR Steel 1006. The pulley setup supports dual operational modes (pull-up and pull-down) and is fixed securely using four L-brackets.

B. Design for Manufacturing

The selected materials—HR Steel 1006, ASTM 20 gray cast iron, and aluminum—are readily available and compatible with standard manufacturing processes. Key processes include:

- I. Cutting and shaping of the aluminum frame using CNC or manual milling.
- II. Lathe operations for machining the pulley shaft and turning the pulleys.
- III. Drilling to create holes for mounting brackets and pulleys.

All fabrication steps are simple and can be carried out using conventional tools, making the design workshop-friendly.

C. Design for Assembly

- The entire structure is modular, designed for easy assembly with the chosen bolts, nuts of appropriate sizes and standardized fasteners.
- The L-brackets provide structural integration between the guide rail plate and the base frame.
- Dumbbell disks can be manually loaded onto the bar by the user, eliminating the need for dedicated weight stacks.
- The system can be disassembled and reassembled easily, supporting transportability and maintenance.

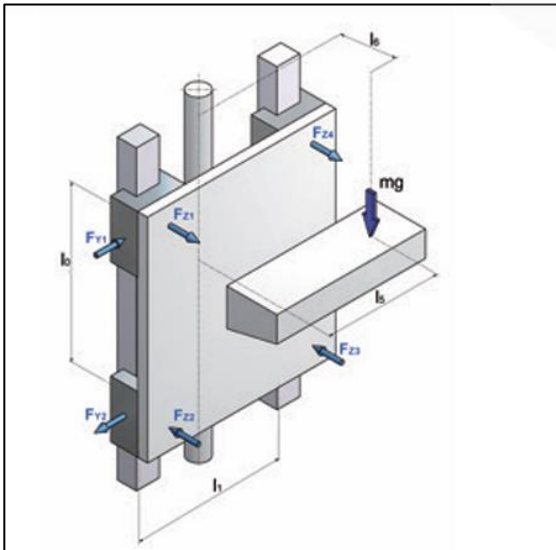
D. Factoring Smoothness in Design

Linear guide rails ensure minimal friction and allow for consistent, stable motion throughout the exercise range. Precision-aligned pulleys reduce resistance during load movement, offering a natural feel for both pull-up and pull-down exercises. Frame design with extended legs improves overall stability and minimizes wobbling under high loads. Proper spacing and material transitions contribute to a smooth user experience and low maintenance needs.

6. Design Calculations

A. Vertical installation of Linear Guide Rails

Given Parameters



$$v = \frac{2}{3} \text{ m/s}$$

$$s_1 = 200 \text{ mm}$$

$$s_2 = 600 \text{ mm}$$

$$s_3 = 200 \text{ mm}$$

$$m = 80 \text{ kg}$$

$$g = 9.8 \text{ m/s}^2$$

$$l_0 = 100 \text{ mm}, \quad l_1 = 100 \text{ mm}, \quad l_5 = 0 \text{ mm}, \quad l_6 = 400 \text{ mm}$$

$$C = 17.71 \text{ kN}$$

$$C_0 = 30.50 \text{ kN}$$

$$f_w = \frac{16}{9}$$

a) Active Loads Per Carriage

During the Acceleration Phase

Radial Loads:

$$F_{\text{Besch}21} = \frac{m(g+a) \cdot l_6}{2 \cdot l_0} = \frac{80 \cdot \left(9.8 + \frac{10}{3}\right) \cdot 400}{2 \cdot 100} = 2102.93 \text{ N}$$

$$F_{\text{Besch}22} = -2102.93 \text{ N}$$

$$F_{\text{Besch}23} = -2102.93 \text{ N}$$

$$F_{\text{Besch}24} = 2102.93 \text{ N}$$

Tangential Loads:

$$F_{\text{Besch}Y1} = F_{\text{Besch}Y2} = F_{\text{Besch}Y3} = F_{\text{Besch}Y4} = 0 \text{ N}$$

At Constant Velocity

Radial Loads:

$$F_{\text{Konst}Z1} = \frac{mg \cdot l_6}{2 \cdot l_0} = \frac{80 \cdot 9.8 \cdot 400}{2 \cdot 100} = 1569.6 \text{ N}$$

$$F_{\text{Konst}Z2} = -1569.6 \text{ N}$$

$$F_{\text{Konst}Z3} = -1569.6 \text{ N}$$

$$F_{\text{Konst}Z4} = 1569.6 \text{ N}$$

Tangential Loads:

$$F_{\text{Konst}Y1} = F_{\text{Konst}Y2} = F_{\text{Konst}Y3} = F_{\text{Konst}Y4} = 0 \text{ N}$$

During the Deceleration Phase

Radial Loads:

$$F_{\text{Verz}Z1} = \frac{m(g-a) \cdot l_6}{2 \cdot l_0} = \frac{80 \cdot \left(9.8 - \frac{10}{3}\right) \cdot 400}{2 \cdot 100} = 1036.26 \text{ N}$$

$$F_{\text{Verz}Z2} = -1036.26 \text{ N}$$

$$F_{\text{Verz}Z3} = -1036.26 \text{ N}$$

$$F_{\text{Verz}Z4} = 1036.26 \text{ N}$$

Tangential Loads:

$$F_{\text{Verz}Y1} = F_{\text{Verz}Y2} = F_{\text{Verz}Y3} = F_{\text{Verz}Y4} = 0 \text{ N}$$

b) Combined Loads per Carriage (Equation [2.8])

Acceleration Phase:

$$F_{\text{Besch}E1} = F_{\text{Besch}E2} = F_{\text{Besch}E3} = F_{\text{Besch}E4} = |2102.93 \text{ N}|$$

Constant Velocity:

$$F_{\text{Konst}E1} = F_{\text{Konst}E2} = F_{\text{Konst}E3} = F_{\text{Konst}E4} = |1569.6 \text{ N}|$$

Deceleration Phase:

$$F_{\text{Verz}E1} = F_{\text{Verz}E2} = F_{\text{Verz}E3} = F_{\text{Verz}E4} = |1036.26 \text{ N}|$$

c) Static Safety Factor

$$f_s = \frac{C_0}{F_{0\text{MAX}}} = \frac{30500}{2102.93} = 14.5$$

d) Dynamic Equivalent Load

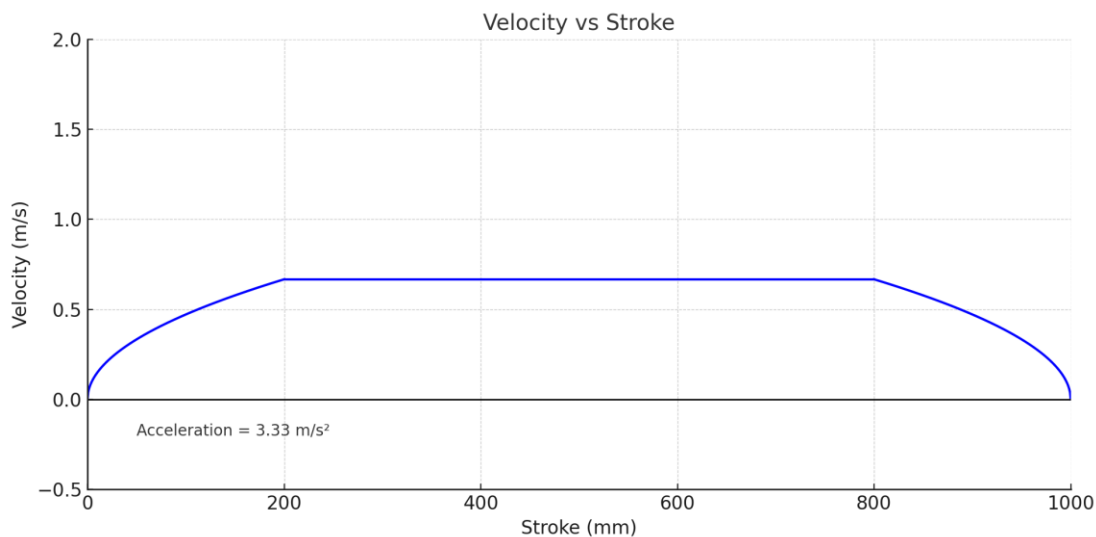
$$S = S_1 + S_2 + S_3 = 1000 \text{ mm}$$

$$\begin{aligned} F_{m1} &= \sqrt[3]{\frac{1}{S} (F_{\text{Besch}E1}^3 \cdot S_1 + F_{\text{Konst}E1}^3 \cdot S_2 + F_{\text{Verz}E1}^3 \cdot S_3)} \\ &= \sqrt[3]{\frac{1}{1000} (2102.93^3 \cdot 200 + 1569.6^3 \cdot 600 + 1036.26^3 \cdot 200)} \\ &= 1638.97 \text{ N} \end{aligned}$$

e) Nominal Life

$$Life_{\text{distance}} = \left(\frac{C}{F_{m1}} \cdot \frac{f_H \cdot f_T \cdot f_C}{f_W} \right)^3 \cdot 5 \cdot 10^4 = \left(\frac{17,710}{1638.97} \cdot \frac{9}{16} \right)^3 \cdot 5 \cdot 10^4 = 11,227.3041 \text{ km}$$

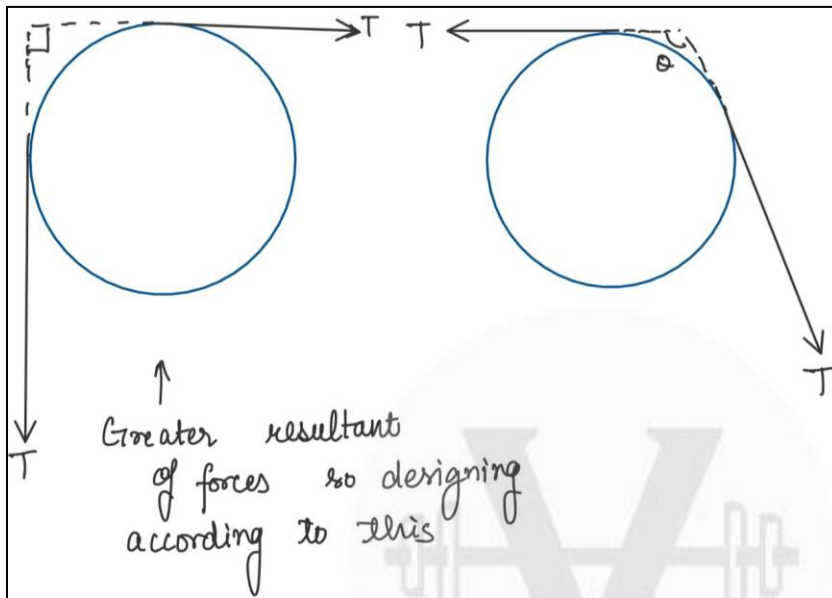
$$\text{Number of Cycles to failure} = \frac{1.122 \times 10^7}{2} = 5.61 \times 10^6$$



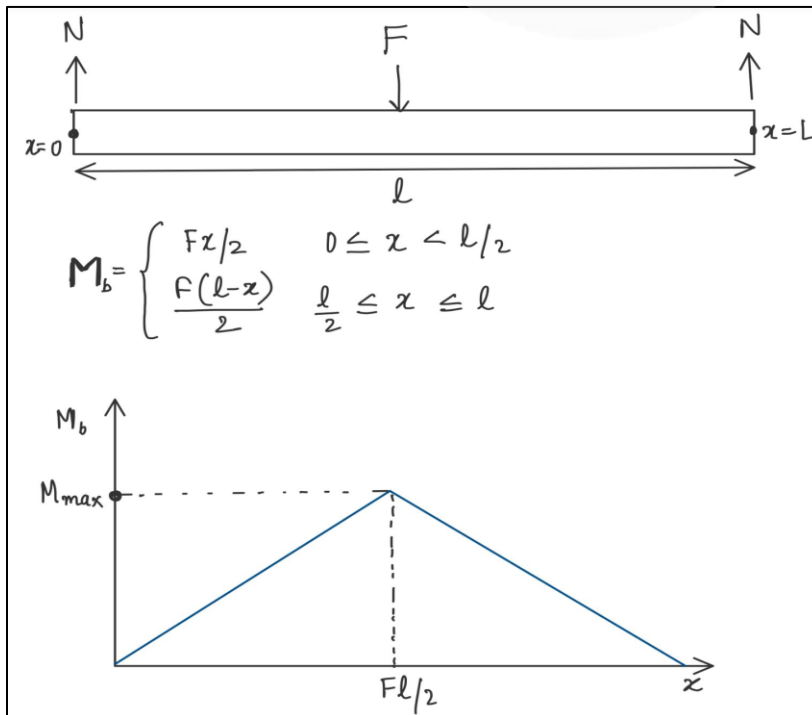
B.

Pulley Calculations (Bearing and Shaft)

FBD for two pulleys at the top:



FBD for shaft through the pulley:



Calculations for Pulley:**Mass:**

Assuming the pulley is a disc of radius $r = 7.5 \text{ cm} = 7.5 \times 10^{-2} \text{ m}$, thickness $t = 2 \times 10^{-2} \text{ m}$, and density $\rho = 7850 \text{ kg/m}^3$, the mass M of the pulley is given by:

$$M = \frac{\pi r^2 t \rho}{4} = \frac{\pi (7.5 \times 10^{-2})^2 (2 \times 10^{-2}) (7850)}{4} = 2.77 \text{ kg}$$

Moment of Inertia:

$$I = \frac{1}{2} M r^2 = \frac{1}{2} \times 2.77 \times (7.5 \times 10^{-2})^2 = 7.8 \times 10^{-3} \text{ kg} \cdot \text{m}^2$$

Force Calculations:

$$T_1 = (m + M)(g + a) = 80 \times (9.8 + 3.33) = 1051.2 \text{ N}$$

Using the relation $R\alpha = a$, we get:

$$\alpha = \frac{a}{r} = \frac{3.33}{7.5 \times 10^{-2}} = 44.4 \text{ rad/s}^2$$

$$T_2 = T_1 + I\alpha = 1051.2 + (7.8 \times 10^{-3}) \times 44.4 = 1051.54632 \text{ N}$$

Bearing Life Calculations:

Assuming a life factor for misalignment safety:

$$f_L = 2$$

$$F_r = \frac{T_2 + T_1}{\sqrt{2}} = \frac{1051.54632 + 1051.2}{\sqrt{2}} = 1487 \text{ N} = 1.487 \text{ kN}$$

Considering an outer rotation factor $V = 1.2$. We check:

$$\frac{f_L \cdot F_a}{C_0} = 0.014 \quad (\text{verysmall})$$

Since:

$$\frac{F_a}{V \cdot F_r} < e$$

we use:

$$F'_e = f_L \cdot V \cdot F_r = 2 \times 1.2 \times 1.487 = 3.5686 \text{ kN}$$

Average RPM:

$$RPM_{avg} = \frac{V_{avg}}{r} \cdot \frac{60}{2\pi} = \frac{0.667}{7.5 \times 10^{-2}} \cdot \frac{60}{2\pi} \approx 86 \text{ RPM}$$

Design Life Requirement:

To ensure a 2-year warranty with safety margin, we design for 3 years:
usage: 4 hours

$$L_h = 365 \times 3 \times 4 = 4380 \text{ hours}$$

$$L_d = 60 \times L_h \times RPM_{avg} = 60 \times 4380 \times 86$$

Required dynamic load rating is given by:

$$C_{10} = F'_e \left(\frac{L_d}{L_{10}} \right)^{1/3}$$

$$C_{10} = 10.06 \text{ kN} \quad (\text{calculated})$$

Chosen bearing:

$$C_{10} = 12.4 \text{ kN}, \quad \text{Bore} = 20 \text{ mm}, \quad \text{Outer Diameter} = 47 \text{ mm}$$

Calculation of Bearing for Bottom Pulley:

This is a smaller pulley. Ignoring its mass (and hence moment of inertia), we approximate:

$$T = (M + m)(g + a) = 1051.2 \text{ N}$$

$$\theta \approx 81.77^\circ$$

$$F_r = 2T \cos \left(\frac{90^\circ - \theta}{2} \right) = 2.097 \text{ kN}$$

Again,

$$\frac{F_a}{C_0} \ll 0.014 \Rightarrow \frac{F_a}{C_0} = 0.014$$

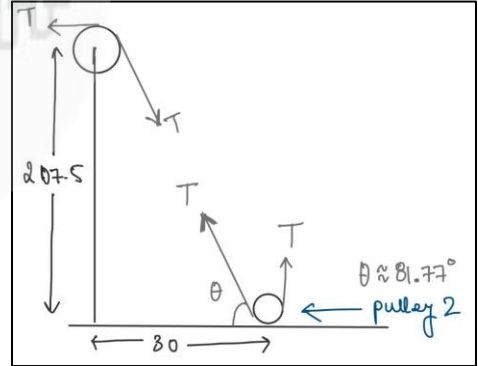
$$F'_e = f_L \cdot V \cdot F_r = 2 \times 1.2 \times 2.097 = 5.0328 \text{ kN}$$

$$L_h = 4380 \text{ hrs} \Rightarrow L_d = 60 \times 4380 \times 86 = 22.42 \times 10^6$$

$$C_{10} = F'_e \left(\frac{L_d}{L_{10}} \right)^{1/3} \approx 13.87 \text{ kN}$$

Chosen bearing:

$$C_{10} = 14 \text{ kN}, \quad \text{Bore} = 25 \text{ mm}, \quad \text{Outer Diameter} = 52 \text{ mm}$$



Calculations for Rods on Top Pulleys

Rod of Pulley 1 (Worst Case)

$$F_1 = T\sqrt{2}, \quad l = 4 \text{ cm}, \quad T = 1051.2 \text{ N}, \quad d = 19.995 \text{ mm}$$

For Yielding:

$$M_{max} = \frac{Fl}{2} = \frac{T\sqrt{2} \cdot l}{2}$$

$$\sigma_b = \frac{32M}{\pi d^3} = \frac{32 \cdot T\sqrt{2} \cdot l}{2\pi d^3} = 37.8849 \text{ MPa}$$

$$S_y = 170 \text{ MPa}, \quad n_f = \frac{S_y}{\sigma_b} = \frac{170}{37.8849} \approx 4.48$$

For Fatigue:

$$\sigma_{b,max} = 37.885 \text{ MPa}, \quad \sigma_{b,min} = 18.683 \text{ MPa}$$

$$\sigma_m = 28.284 \text{ MPa}, \quad \sigma_a = 9.601 \text{ MPa}$$

$$S'_e = 0.5 \cdot S_{ut} = 150 \text{ MPa}$$

$$K_a = 7.7 \cdot (300)^{-0.718} = 0.961$$

$$d' = 0.368 \cdot d = 7.7181 \text{ mm}, \quad K_b = 1.24 \cdot (d')^{-0.107} = 0.9965$$

$$K_c = 1, \quad K_d = 1, \quad K_e = 0.814, \quad K_f = 1$$

$$S_e = K_a K_b K_c K_d K_e K_f \cdot S'_e = 116.93 \text{ MPa}$$

$$n_f = \left(\frac{\sigma_a}{S_e} + \frac{\sigma_m}{S_{ut}} \right)^{-1} = 5.6693$$

Rod of Pulley 2 (Bottom Pulley)

$$F = 2T \cos\left(\frac{180 - \theta}{2}\right) = 2 \cdot 1051.2 \cdot \cos\left(\frac{99}{2}\right) = 2095.919 \text{ N}$$

$$l = 4 \text{ cm}, \quad d = 24.993 \text{ mm}, \quad M = \frac{Fl}{2} = 41.91838 \text{ Nm}$$

$$\sigma_b = \frac{32M}{\pi d^3} = 27.3495 \text{ MPa}$$

$$n_f = \frac{S_y}{\sigma_b} = \frac{170}{27.3495} = 6.2158$$

Fatigue:

$$\sigma_{b,max} = 27.3495 \text{ MPa}, \quad \sigma_{b,min} = 13.4696 \text{ MPa}$$

$$\sigma_m = 20.409 \text{ MPa}, \quad \sigma_a = 6.94 \text{ MPa}$$

$$d' = 0.368 \cdot 24.993 = 9.1974 \text{ mm}, \quad K_b = 1.24 \cdot (d')^{-0.107} = 0.9779$$

$$S_e = 114.7449 \text{ MPa}, \quad n_f = \left(\frac{\sigma_a}{S_e} + \frac{\sigma_m}{S_{ut}} \right)^{-1} = 7.7814$$

C. Tension Rope Calculation

Calculations for Rope

Fatigue Failure

$$F_{max} = (m + M)(g + a), \quad F_{min} = (m + M)(g - a)$$

$$d = 6 \text{ mm}, \quad \sigma_{max} = 37.17 \text{ MPa}, \quad \sigma_{min} = 18.325 \text{ MPa}$$

$$\sigma_a = \frac{\sigma_{max} - \sigma_{min}}{2} = 9.4255 \text{ MPa}, \quad \sigma_m = \frac{\sigma_{max} + \sigma_{min}}{2} = 27.75 \text{ MPa}$$

$$S'_e = 700 \text{ MPa}, \quad S_{ut} = 1770 \text{ MPa}$$

Modification Factors

$$K_a = a \cdot S_{ut}^b = 4.51 \cdot (1770)^{-0.265} \approx 0.6215$$

$$K_b = 1 \text{ (axial loading)}, \quad K_c = 0.85, \quad K_d = 1, \quad K_e = 0.814, \quad K_f = 1$$

$$S_e = K_a K_b K_c K_d K_e K_f \cdot S'_e = 0.6215 \cdot 1 \cdot 0.85 \cdot 1 \cdot 0.814 \cdot 1 \cdot 700 = 301.026 \text{ MPa}$$

$$n_f = \left(\frac{\sigma_a}{S_e} + \frac{\sigma_m}{S_{ut}} \right)^{-1} = \left(\frac{9.4255}{301.026} + \frac{27.75}{1770} \right)^{-1} = 21.28$$

Yielding

$$S_y = 1240 \text{ MPa}, \quad n_f = \frac{S_y}{\sigma_{max}} = \frac{1240}{37.17} = 33.358$$

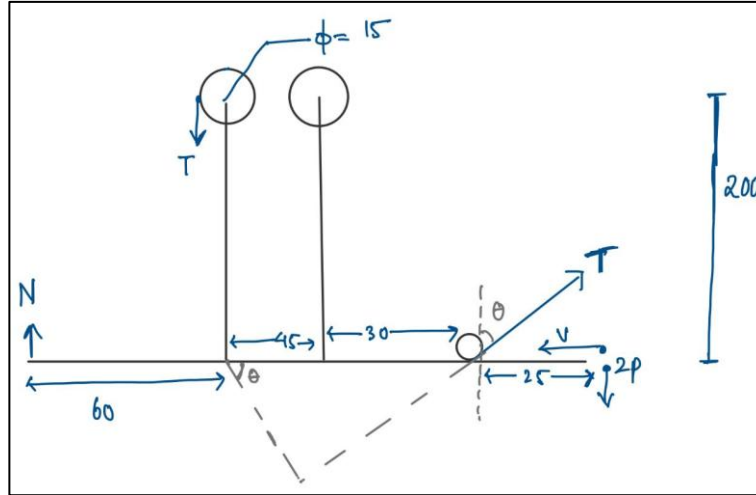
Conclusion:

Material: Cold Drawn Steel Core PVC coated wire

D. Bolted Joints Calculations

Calculations for Bolt

FBD for the system in observation:



Here, we have assumed that the bolts are present to give support to the base frame, or else it may topple about the other end.

Thus, normal force will shift and the end and external load on each of two bolts ahead is assumed to be P

$$T \cos \theta - T + N - 2P = 0 \Rightarrow N = 2P + T(1 - \cos \theta) \rightarrow \text{Force Balance}$$

$$T(7.5 + 75 \cos \theta) = 200P + 60N = 200P + 120P + 60T - 60T \cos \theta \rightarrow \text{Moment Balance}$$

$$T(135 \cos \theta - 52.5) = 320P \Rightarrow P = \frac{T(135 \cos \theta - 52.5)}{320}$$

$$V = T \sin \theta$$

Operating range: $0^\circ \leq \theta \leq 45^\circ$ According to NASA [V], for combined loading in bolts:

$$\left(\frac{V}{V_{\text{allow}}} \right)^2 + \left(\frac{P}{P_{\text{allow}}} \right)^2 = \frac{1}{n_f^2} \leq 1$$

$$V_{\text{allow}} = \frac{2}{3} S_{ut} \cdot A_r, \quad P_{\text{allow}} = S_{ut} \cdot A_t$$

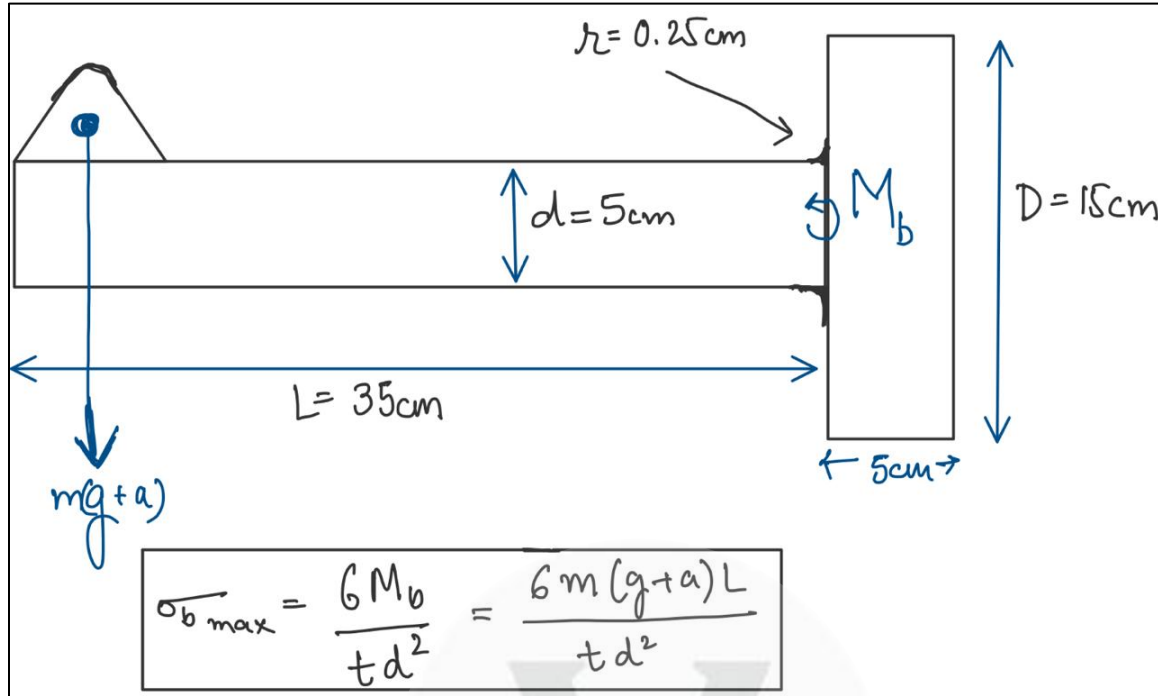
Chosen: M5 grade 4.8 ($A_t = 14.2 \text{ mm}^2$, $A_r = 12.7 \text{ mm}^2$, $S_{ut} = 420 \text{ MPa}$)
Graphically, we observe that n_f is minimum at $\theta = 67^\circ$ and thus our operating range is well below the point with maximum loading conditions.

Using $T = M(g + a) = 1051.2 \text{ N}$, we get **$\min(n_f) = 3.675$**

We will place 4 of such bolts on the base frame, one at each corner to provide additional safety against toppling of the equipment.

E. Weight Holder Component Calculations

a) Against Fracture



$m(g+a)L = M$ at critical point
applying C.M.

$$\frac{\sigma'_1}{S_{ut}} - \frac{\sigma'_2}{S_{uc}} = \frac{1}{\eta_d} < 1$$

$$\sigma_1 = -\sigma_2 = \frac{6m(g+a)L}{td^2} \Rightarrow \sigma'_1 = -\sigma'_2 = \frac{6m(g+a)Lk_t}{td^2}$$

finally

$$\eta_{pos} = \left(\frac{S_{ut} \cdot S_{uc}}{S_{ut} + S_{uc}} \right) \frac{td^2}{6m(g+a)LK_t}$$

Given:

$$\begin{aligned} D &= 15\text{ cm} & \frac{D}{d} &= 3 \\ d &= 5\text{ cm} & t &= 10\text{ cm} \\ \frac{r}{d} &= 0.05 & m &= 80\text{ kg} \\ L &= 35\text{ cm} & K_t &= 2.4 \end{aligned}$$

ASTM 20 cast iron:

$$\begin{aligned} S_{ut} &= 152\text{ MPa} \\ S_{uc} &= 572\text{ MPa} \end{aligned}$$

Therefore, $n_{fos} = 5.665$ is the factor of safety against fracture.

b) Fatigue

Fatigue Failure Calculations

$$\sigma_{b \max} = \frac{6M_b}{td^2} = \frac{6m(g+a)L}{td^2}$$

$$\sigma_{b \min} = \frac{6m(g-a)L}{td^2}$$

$$\sigma_m = \frac{6mgL}{td^2} = 6.592 \text{ MPa}$$

$$\sigma_a = \frac{6maL}{td^2} = 2.24 \text{ MPa}$$

$$S'_c = 0.5S_{ut} = 0.5(152) = 76 \text{ MPa}$$

$$k_a = 1$$

$$\text{For } k_b : d' = \sqrt{0.652dt} = 57.096 \text{ mm}$$

$$\text{Thus, } k_b = 1.51(d')^{-0.157} = 0.8$$

$$k_c = 1, \quad k_d = 1, \quad k_e = 0.814, \quad k_f = 1, \quad K_f = 1$$

$$S_e = k_a k_b k_c k_d k_e k_f S'_c$$

$$= 49.4912 \text{ MPa}$$

$$n_f = \left(\frac{\sigma_a}{S_e} + \frac{\sigma_m}{S_{ut}} \right)^{-1}$$

$$= 11.283$$

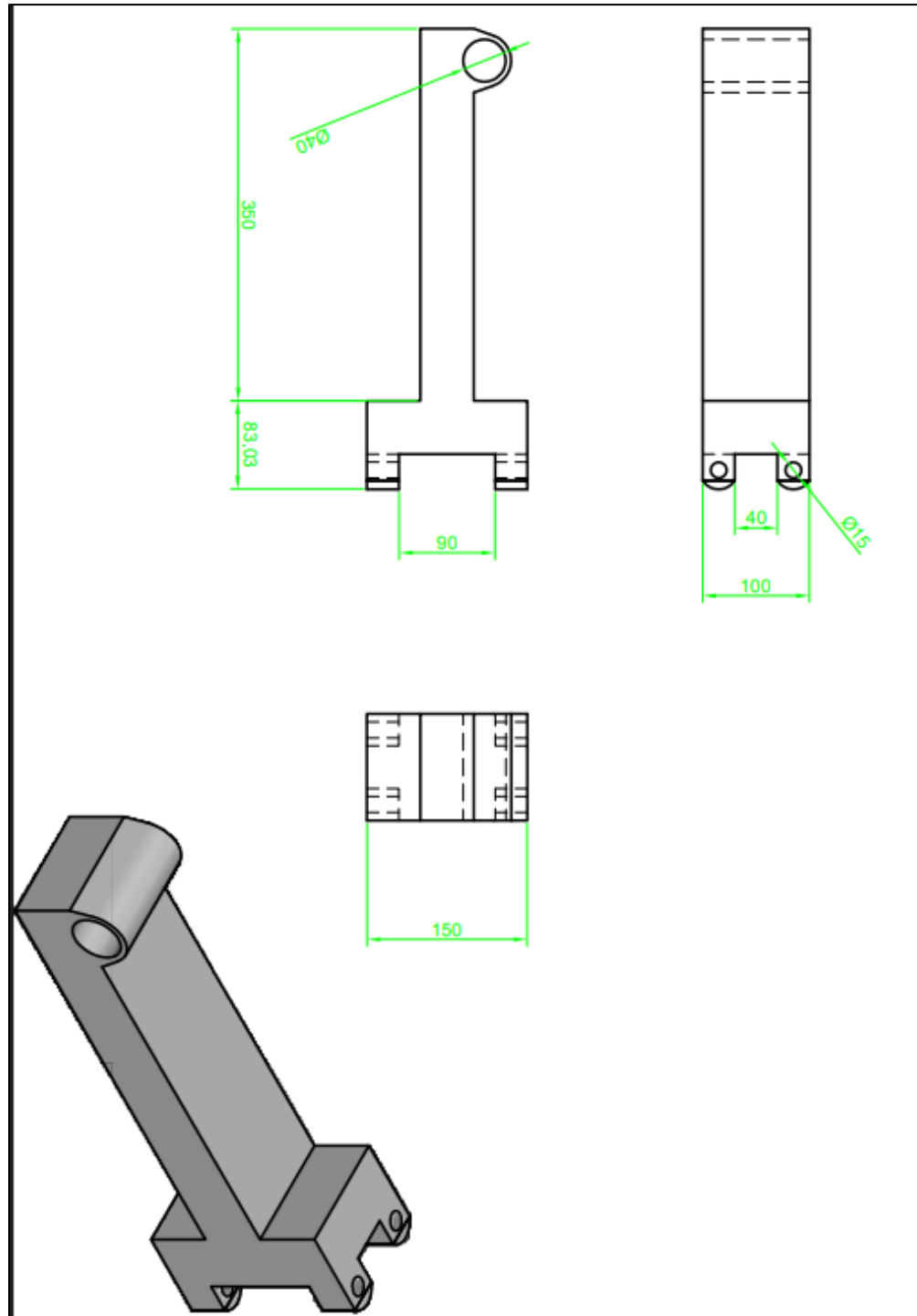
Results based on calculation:

Type Of Failure	Factor of Safety	Chosen Material
Linear Guide Rail	14.5	LGBCH20FN, 2-Carriage per rail
Pulley Bearing	-	a) Bore=20mm (d _{pulley} =15cm) b) Bore=25mm (d _{pulley} =10cm)
Pulley Shafts	(i) n _y =4.48 n _f =5.67 (ii) n _y = 6.22 n _f = 7.78	AISI 1006 HR Steel
Rope	n _y = 33.36 n _f = 21.28	CD Steel core with PVC coating
Bolt	3.675	Grade 4.8 M5
Weight Holder	n _{fracture} = 5.67 n _{fatigue} = 11.28	ASTM 20 cast Iron

7. CAD Models and Drawings

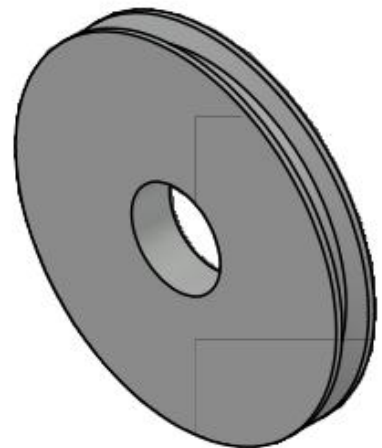
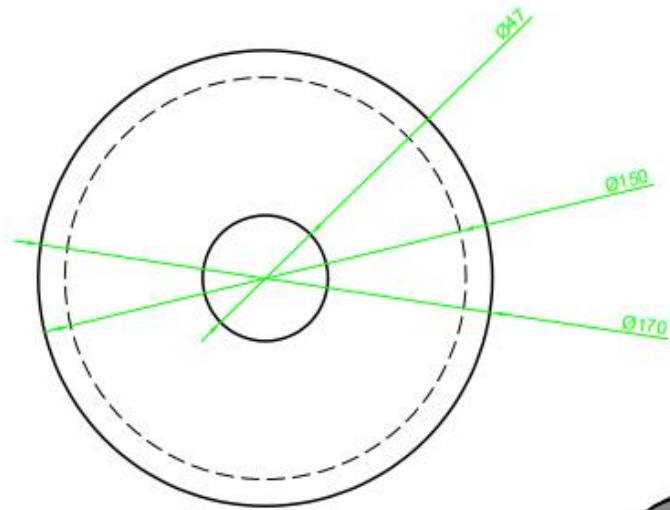
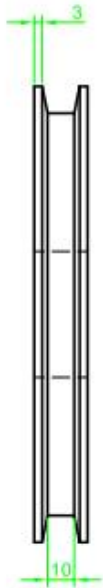
Part Drawings

*All dimensions
are in mm*

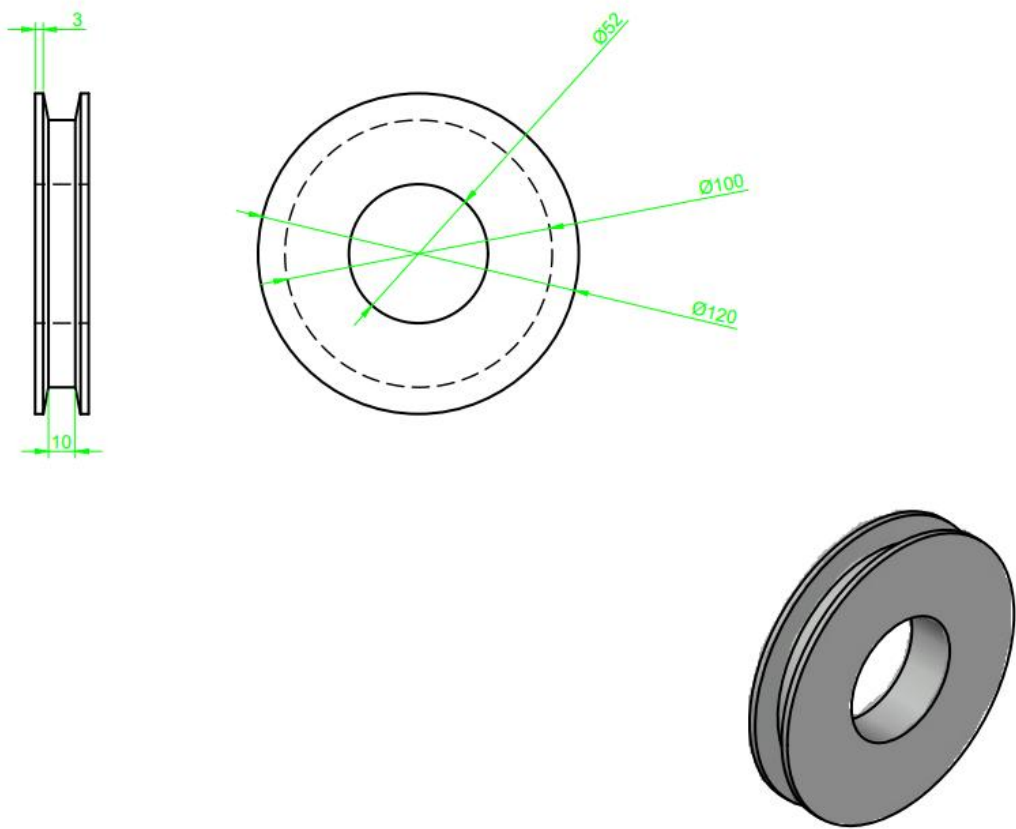


Weight Holder (Without shaft)

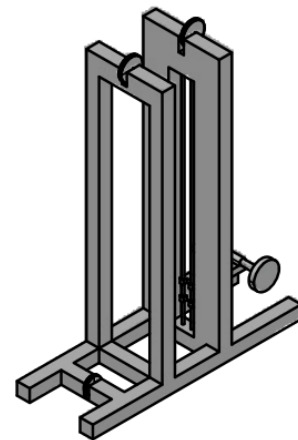
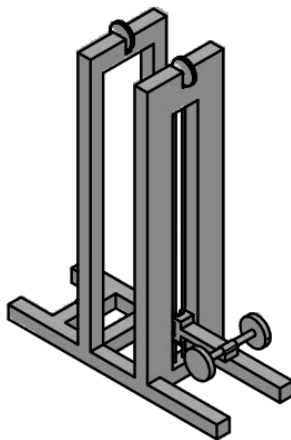
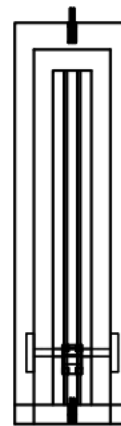
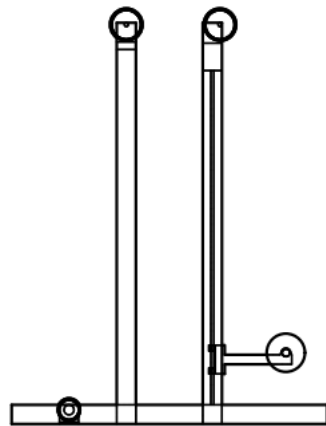
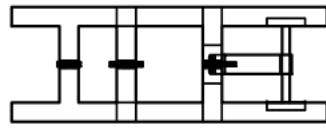
Upper Pulley



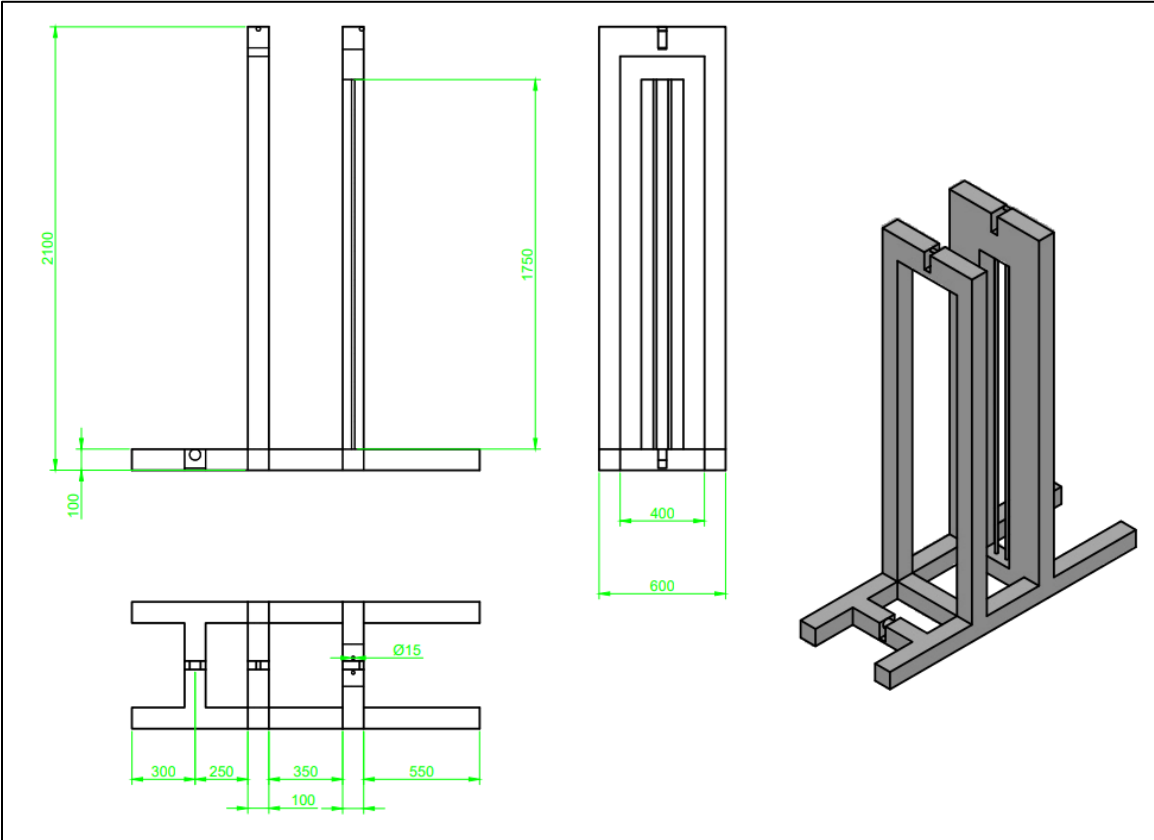
Lower Pulley



Assembly Drawing



Assembly Drawing of product
without the cable



Frame

Bill Of Materials

SNo.	Name	Material	Quantity
11	Pulley (d = 15 cm)	Stainless Steel	2
10	Pulley (d = 10 cm)	Stainless Steel	1
9	Frame	Aluminium	1
8	Linear Guide Rail	LGBCH20FN	2 x 2
7	Weight holder	ASTM 20 Cast Iron	1
6	Shaft (L = 4 cm; D = 20 mm)	AISI 1006 HR Steel	2
5	Shaft (L = 4 cm; D = 25 mm)	AISI 1006 HR Steel	1
4	Shaft (L = 40 cm; D = 40 mm)	AISI 1006 HR steel	1
3	Bearing (Bore = 20 mm)	-	2
2	Bearing (Bore = 25 mm)	-	1
1	Bolts	Grade 4.8 M5	4

8. Prototyping and Testing Plan

A. Prototyping Strategy

To validate the design concept and assess functionality, a working prototype of the gym equipment will be developed. The prototype will include all key mechanical components such as:

- Linear guide rail system
- Pulley mechanism (top and bottom)
- Dumbbell-loadable bar
- Base frame with extended supports
- Mounting brackets and shaft holders
- Nuts, bolts and bearings

Materials and Fabrication Steps:

Frame Fabrication: Aluminum sections will be cut and drilled to match the base design. Extension arms will be bolted or welded as required.

Guide Rail and Carriage: The 2-meter-long rail will be mounted on the frame using L-brackets, with the cast iron carriage installed to move along it. This will be welded and connected to the base frame.

Pulley System: Pulleys will be attached to the HR Steel 1006 shaft, which is inserted into drilled holes at the top of the guide plate.

Bar Attachment: The steel bar will be mounted to the carriage, with proper load brackets to hold dumbbell disks.

Assembly: All parts will be assembled using standard fasteners. Alignment and spacing will be verified to ensure proper motion.

B. Testing Objectives

The prototype will be tested for mechanical integrity, usability, and smoothness of motion. The goals of testing include:

- ✓ **Load Testing:** Assess maximum safe load using standard dumbbell disks.
- ✓ **Motion Testing:** Check for friction, resistance, and smooth travel of the carriage along the guide rail.
- ✓ **Switching Modes:** Evaluate ease and reliability of switching between pull-down and pull-up modes.
- ✓ **Durability Test:** Simulate repetitive motion cycles to assess wear and potential failure points.

- ✓ **Stability Check:** Confirm base support and balance under asymmetric and high-load conditions.

C. Evaluation Criteria

- ✓ **Smoothness of operation:** Minimal jerks, consistent motion.
- ✓ **Load capacity:** Ability to handle expected user weights without deformation or instability.
- ✓ **Safety:** No sharp edges, pinch points, or loose components.
- ✓ **Ease of use:** Quick loading/unloading of weights, effortless mode switching.
- ✓ **Maintainability:** Accessibility of parts for tightening or replacement.

D. Feedback and Iteration

Results from testing will be recorded and compared against functional requirements. Any design flaws or limitations will be addressed through iterative improvements. Potential upgrades (e.g., adjustable handles, integrated sensors) may be considered based on user feedback.

9. Future Scope and Improvements

While the current design meets the core functional and structural goals of a dual-mode resistance training machine, several enhancements can be explored to increase usability, functionality, and user engagement:

1. Integration of a Digital Rep Counter

A digital rep counter can be added to automatically track the number of repetitions completed by the user. This can be implemented using:

- ✚ Reed switches or limit switches on the carriage to count one rep each time it reaches the bottom or top end.
- ✚ A microcontroller (e.g., Arduino or ESP32) with an LCD display to show the rep count in real time.
- ✚ Optional reset and memory functions to track sets and overall workout progress.

2. Smart Speed Sensor for Form Feedback

The quality of a workout greatly depends on performing repetitions at the correct speed and rhythm. Repetitions that are too fast or too slow can compromise form, reduce effectiveness, and increase injury risk. A sensor-based system can be added to monitor motion speed and provide live feedback. Linear motion sensor (e.g., rotary encoder or infrared distance sensor) mounted along the guide rail to track the movement of the carriage or weight. Alternatively, pulley rotation sensors (using a rotary encoder or hall effect sensor) can measure the speed of the rotating pulley and infer motion speed. Based on collected data, the system can:

- ✚ Calculate average and peak speed of each rep.
- ✚ Alert the user (via a buzzer, vibration, or LED indicator) if reps are too fast or too slow.
- ✚ Log motion data for post-workout analysis on a mobile app or PC.

3. Mobile App Integration

Bluetooth/Wi-Fi modules can be added to sync data with a mobile application. The app can provide visual feedback, rep history, speed/form charts, and recommendations for improvement.

4. Additional Attachments and Adjustability

- ✚ Interchangeable handles or grips for varied exercises (e.g., triceps rope, lat bar).
- ✚ Height-adjustable pulley anchors to expand exercise options (e.g., rows, curls).
- ✚ Foldable frame design for improved portability and storage.

5. Safety and Ergonomics Upgrades

Soft padding or bumpers at end-stops to prevent impact damage. Automatic locking mechanism for emergency stop or carriage holding in case of cable failure.

10. Conclusion

The development of the ClevBell gym equipment represents a blend of simplicity, functionality, and adaptability in fitness technology. Designed with a dual-mode operation—supporting both pull-down and pull-up resistance exercises—the system makes effective use of linear guide rails and a loadable bar to offer a customizable workout experience using standard dumbbell plates.

Material selection, frame geometry, and modular integration have been carefully considered to ensure the structure is lightweight, stable, and user-friendly. The use of HR Steel 1006, ASTM 20 gray cast iron, and aluminum balances strength with ease of manufacturing.

With an eye on innovation, the design also leaves room for future expansion, such as integrating smart sensors for rep tracking and form evaluation. These upgrades not only enhance user engagement but also support better workout outcomes through real-time feedback.

This product is a strong candidate for low-cost, portable resistance training—ideal for individuals seeking an efficient home workout solution or compact gym equipment. Further testing, iterations, and digital enhancements will help refine the product into a market-ready fitness innovation.

11. References

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